

7. Interface

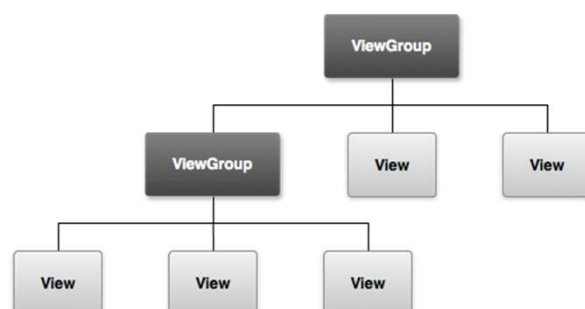
30-set-2025

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Layout

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- Os elementos de interface com o utilizador são conseguidos através da utilização de objetos **View** e **ViewGroup**
- View - desenha um elemento de interação
- ViewGroup - contém ou agrupa objetos do tipo View

<https://developer.android.com/guide/topics/ui/declaring-layout>

Tipos de layout

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The diagram illustrates six common Android layout types, each represented by a rounded rectangle icon:

- LinearLayout**: Shows three rectangular boxes arranged horizontally.
- RelativeLayout**: Shows a large rectangle with a smaller one inside it, and another rectangle below the large one.
- FrameLayout**: Shows a single large rectangular box.
- TableLayout**: Shows a table-like structure with three rows and three columns.
- ListLayout**: Shows three horizontal rectangular boxes stacked vertically, with a double-headed vertical arrow to the right.
- GridLayout**: Shows a grid of four columns and three rows of small squares, with a double-headed vertical arrow to the right.

Tipos de layout

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The screenshot shows the Android Studio interface. On the left, the **Layout** palette is open, displaying various layout types. In the center, a visual representation of a **ConstraintLayout** is shown on a dark blue background, featuring a central white circle with arrows indicating constraints to the edges and other views.

developers 

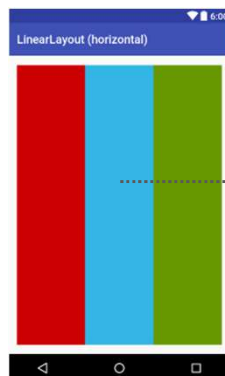
- [ConstraintLayout](#)
- [Build a Responsive UI with ConstraintLayout](#)

Tipos de layout

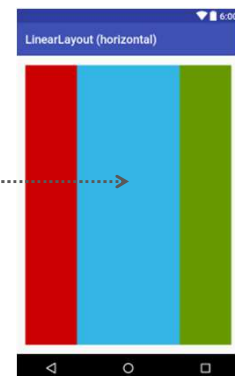
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Layouts

- ConstraintLayout
- GridLayout
- FrameLayout
- LinearLayout (horizontal)
- LinearLayout (vertical)
- RelativeLayout
- TableLayout
- TableRow
- <fragment>



layout_weight="2"

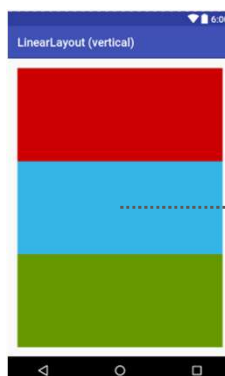


Tipos de layout

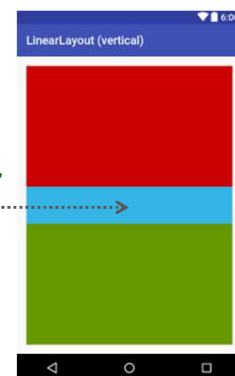
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Layouts

- ConstraintLayout
- GridLayout
- FrameLayout
- LinearLayout (horizontal)
- LinearLayout (vertical)
- RelativeLayout
- TableLayout
- TableRow
- <fragment>



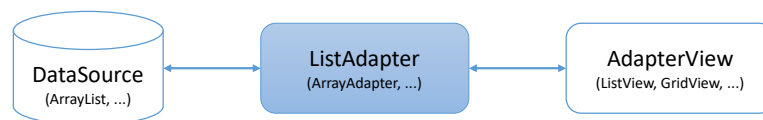
layout_height="..."



ListActivity

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- simple_list_item_1.xml
- simple_list_item_2.xml
- simple_list_item_checked.xml
- simple_list_item_multiple_choice.xml
- simple_list_item_single_choice.xml



adaptado de: Queirós, 2016

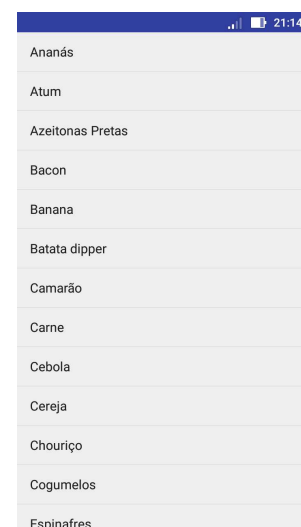
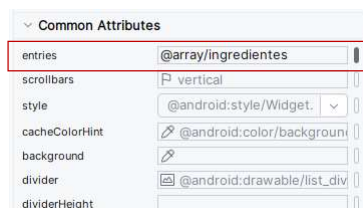
ListActivity

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- res » values » strings.xml

```

<resources>
  <string name="app_name">Pizza</string>
  <string-array name="ingredientes">
    <item>Ananás</item>
    <item>Atum</item>
    <item>Azeitonas Pretas</item>
    <item>Bacon</item>
    <item>Banana</item>
    ...
  
```



ListActivity

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■ kotlin » MainActivity

```
val ingredientes = resources.getStringArray(R.array.ingredientes)
val adapter = ArrayAdapter(
    this,
    android.R.layout.simple_list_item_1,
    ingredientes)
binding.listView.adapter = adapter
```

<https://developer.android.com/guide/topics/ui/layout/listview.html>

ListActivity

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simple_list_item_1

Ananás
Atum
Azeitonas Pretas
Bacon
Banana
Batata dipper
Camarão
Carne
Cebola
Cereja
Chouriço
Cogumelos
Espinafres

simple_list_item_checked

Ananás	✓
Atum	✓
Azeitonas Pretas	✓
Bacon	✓
Banana	✓
Batata dipper	✓
Camarão	✓
Carne	✓
Cebola	✓
Cereja	✓
Chouriço	✓
Cogumelos	✓
Espinafres	✓

simple_list_item_multiple_choice

Ananás	☑
Atum	☐
Azeitonas Pretas	☐
Bacon	☑
Banana	☐
Batata dipper	☐
Camarão	☐
Carne	☐
Cebola	☑
Cereja	☐
Chouriço	☑
Cogumelos	☐
Espinafres	☐

ListView vs RecyclerView

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ListView

RecyclerView

RecyclerView

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- **ViewHolder**
 - Permite um deslocamento suave da lista
 - `onCreateViewHolder()` → criação da view
 - `onBindViewHolder()` → atualização da view

Rui Costa
+351 912 345 678

Classe RecyclerView

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▪ **LayoutManager**

- Responsável pela disposição das linhas (*row views*)
- Permite escolher a forma como se pretende apresentar as linhas e o deslocamento
- LinearLayoutManager → deslocamento vertical e horizontal
- GridLayoutManager → disposição em forma de tabela

Classe RecyclerView

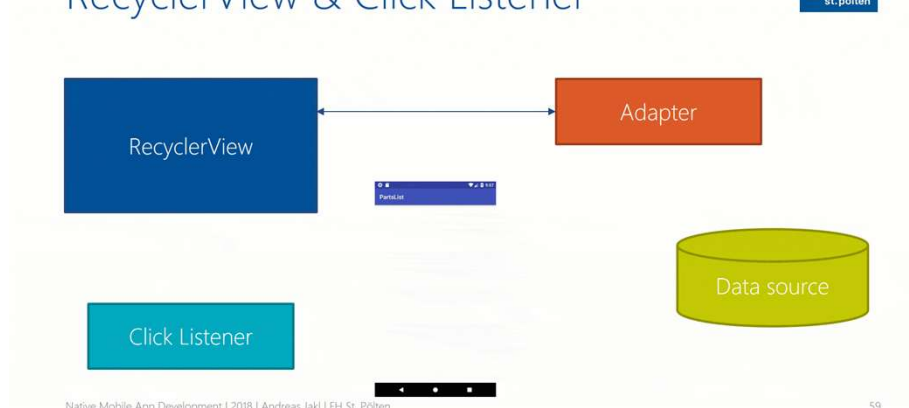
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- <https://developer.android.com/guide/topics/ui/layout/recyclerview>
- What is the Difference Between ListView and RecyclerView?
url: <https://www.thedroidsonroids.com/blog/what-is-the-difference-between-listview-recyclerview>
- What is the difference between ListView and RecyclerView?
url: <https://medium.com/@manuaravindpta/what-is-the-difference-between-listview-and-recyclerview-bcd82c64ffbb>

Classe RecyclerView

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RecyclerView & Click Listener



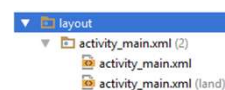
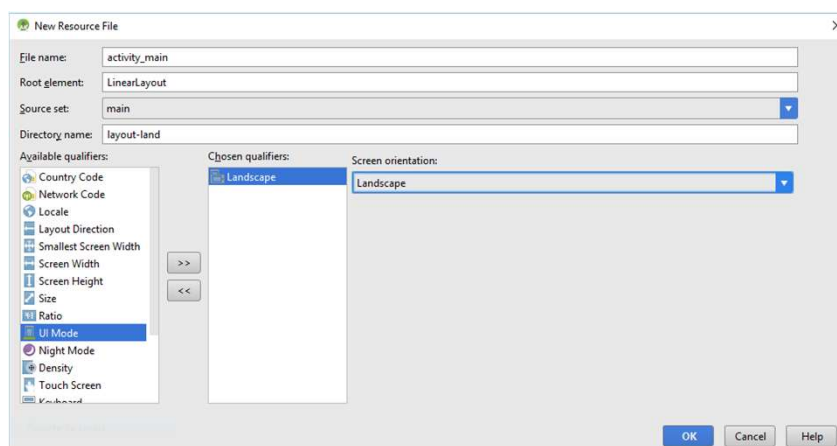
Native Mobile App Development | 2018 | Andreas Jaki | FH St. Pölten

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Qualificadores

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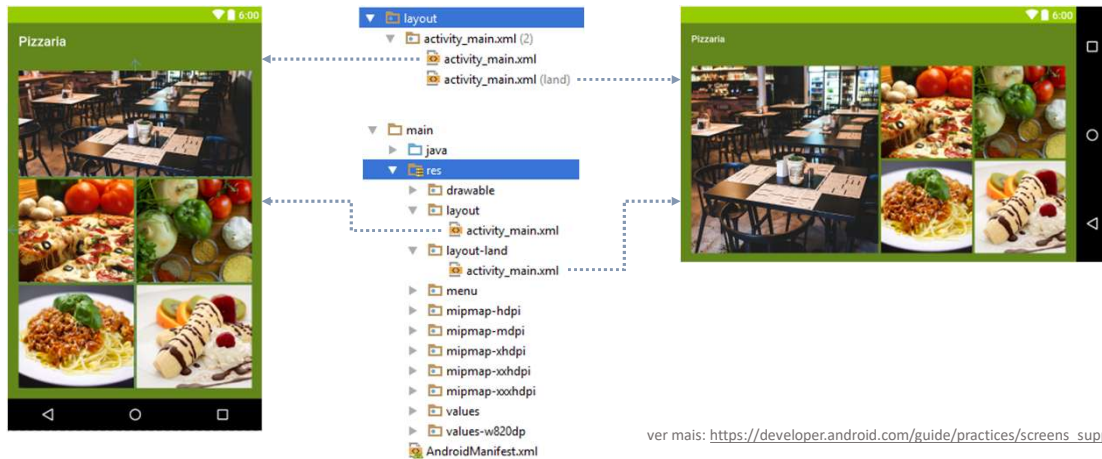
- res » layout » activity_main.xml



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■ res » layout » activity_main.xml



ver mais: https://developer.android.com/guide/practices/screens_support.html